

System Design (Telusko) - Day 1

One-Shot 5h course | Video 0:00-0:15

What is System Design

Understand what system design is, why it matters for large-scale apps, and the difference between functional and non-functional requirements.

Key concepts

- System design = deciding the architecture, components and data flow so an app scales, stays reliable and is maintainable.
- Functional requirements: WHAT the system does (features).
- Non-functional: HOW well - scalability, availability, latency, consistency, durability.
- Always start by clarifying requirements, scale (users/QPS), and constraints.

Key example

Estimate scale first: 1M daily users, ~10 reads : 1 write, peak QPS = daily/86400 * peak factor.

Today's tasks (1.5h)

- [] Watch the intro segment
- [] List functional vs non-functional needs for an app you use
- [] Note the key non-functional trade-offs

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 0:00-0:15)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)